1. Integers from 1 to n.

```
21 ?- consult(['c:/emu8086/integer.pl']).
true.

22 ?- sum(5, S).
S = 15 .
```

2. DB WITH NAME, DOB.

```
24 ?- display(Name, DOB).
Name = arun,
DOB = '2002-05-10' .
```

3. STUDENT-TEACHER-SUB-CODE.

```
39 ?- details(101, Name, Teacher, Code, Subject).
Name = arun,
Teacher = john,
Code = cs101,
Subject = ai.
```

4. PLANETS DB.

```
40 ?- consult(['c:/emu8086/planets.pl']).
true.

41 ?-
|    solar_system(X).
X = mercury
```

5. Towers of Hanoi.

```
25 ?- consult(['c:/Users/rahim/Downloads/bird.pl']).
true.

26 ?- check(parrot).
parrot can fly.
true .

27 ?- check(penguin).
penguin cannot fly.
true.

28 ?- check(sparrow).
sparrow can fly.
true
```

6. bird can fly or not.

```
23 ?- consult(['c:/Users/rahim/Downloads/hanoi.pl']).
true.

24 ?- hanoi(3,a,c,b).
Move disk from a to c
Move disk from a to b
Move disk from c to b
Move disk from a to c
Move disk from b to a
Move disk from b to c
Move disk from a to c
true
```

7. implement family tree.

```
29 ?- consult(['c:/Users/rahim/Downloads/male.pl']).
true.

30 ?- father(ram,raj).
true.

31 ?- mother(sita,raj).
false.

32 ?- sibling(raj,sita).
true.
```

8. Dieting System based on Disease.

```
33 ?- consult(['c:/Users/rahim/Downloads/diet.pl']).
true.

34 ?- suggest(diabetes).
Suggested diet: Low sugar food
true.
```

9. Monkey Banana Problem

```
35 ?- consult(['c:/Users/rahim/Downloads/move.pl']).
true.

36 ?- can_get_banana.
Monkey gets the banana
true.
```

10. color using Back Tracking.

```
37 ?- consult(['c:/Users/rahim/Downloads/fruit.pl']).
true.

38 ?- fruit_color(X,Y).
X = apple,
Y = red ;
X = banana,
Y = yellow ;
X = grapes,
Y = green ;
X = Y, Y = orange.
```

11. Best First Search algorithm

```
39 ?- consult(['c:/Users/rahim/Downloads/edge(a, b, 1).pl']).
true.

40 ?- bestfirst(a,d).
Goal Found: d
true .
```

12. Medical Diagnosis

```
41 ?- consult(['c:/Users/rahim/Downloads/symptom(ravi, fever).pl']).
true.

42 ?- diagnose(ravi).
Disease: flu
true .

43 ?- diagnose(anu).
Disease: headache_problem
true.

44 ?- diagnose(ram).
false.
```

13 forwad Incorporate required queries.

```
45 ?- consult(['c:/Users/rahim/Downloads/forward.pl']).
true.

46 ?- rule1.
Ram is mortal
true.
```

14. backward Chaining. Incorporate required queries.

```
47 ?- consult(['c:/Users/rahim/Downloads/human(ram).pl']).
true.

48 ?- mortal(ram).
true.
```